Electromagnetic Induction

Building Models to Describe our World

Faraday’s Law and Lenz’s Law, and Real World Examples

Outline

Faraday’s Law

Building Models to Describe our World

Faraday’s Law and Lenz’s Law, and Real World Examples

Electromagnetism

- So far, we have looked at static cases:
    1. An electric field that does not change with time (e.g. created by a charge that is not moving)
    2. A magnetic field that does not change with time (e.g. created by a current that is constant)
- We have seen three of the four equations that explain all of electromagnetic phenomena

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I^{enc}$$

Electromagnetism

- So far, we have looked at static cases:
  1. An electric field that does not change with time (e.g. created by a charge that is not moving)
  2. A magnetic field that does not change with time (e.g. created by a current that is constant)

$$\oint \vec{E} \cdot d\vec{A} = \frac{Qenc}{\epsilon_0}$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I^{enc}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

- Faraday's law (the 4th equation) connects electric and magnetic fields
- Faraday's law is the only one that explicitly references time

Maxwell's equations (more or less...)

Faraday's Law

- The circulation of the electric field:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

- This is zero, if $\frac{d\Phi_B}{dt} = 0$, i.e. if Φ_B does not change with time (**statics**).

Faraday's Law

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- This is zero, if $\frac{d\Phi_B}{dt} = 0$, i.e. if Φ_B does not change with time (**statics**).
- What is $\oint \vec{E} \cdot d\vec{l}$?
  - Compare with the definition of electric potential difference:

$$\Delta V = (V_2 - V_1) = -\int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r}$$

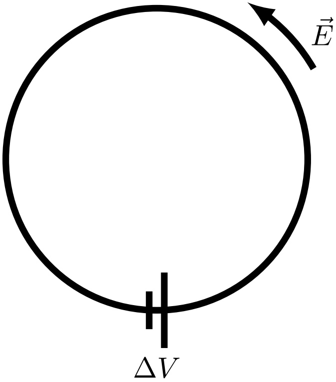
  - $\oint \vec{E} \cdot d\vec{l}$ is the electric potential difference in going around a closed loop (ending where you started)
  - if Φ_B changes with time, then the electric field does net work when you go around a closed path $\rightarrow$ The electric force is no longer conservative!
  - Since ΔV in this case is the potential difference between two points that are the same, we (they) find it less awkward to call it emf (electromotive force).

EMF on a closed loop

- If a charge q goes around the loop, it will gain energy given by:

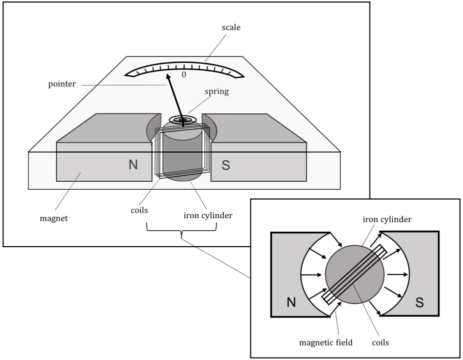
$$W = q\Delta V$$

- If there is no way for the charge to loose energy (e.g. a resistor or air molecules), it will keep going around the circle faster and faster.
- The loop does not need to be an actual physical wire, it can just be an arbitrarily drawn path.
- If the loop is a wire, then the emf is like an ideal battery.



Review: the galvanometer

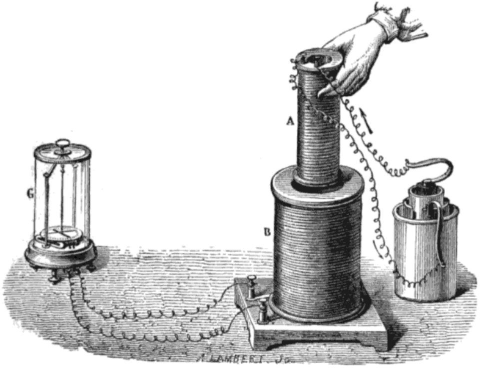
- A coil carrying current has a magnetic dipole moment.
- The coil wants to align its dipole moment with an external magnetic field.
- A galvanometer is a very sensitive device for measuring current:
    - When a current is present in the coil, the torque on the coil is proportional to current.
    - A balancing torque is provided by a restoring spring.



A galvanometer

Induction

- Henry and Faraday both observed that currents can be induced with a magnet.
- Faraday described it in more detail, so we refer to Faraday's Law of Induction.
- Currents were only induced when there was a change in the magnetic field.



Faraday's experiment showing induction between coils of wire: The liquid battery (right) provides a current which flows through the small coil (A), creating a magnetic field. When the coils are stationary, no current is induced. But when the small coil is moved in or out of the large coil (B), the magnetic flux through the large coil changes, inducing a current which is detected by the galvanometer (G). Wikipedia.

More observations from Faraday

- The induced current is larger if the magnetic field is stronger.
- The induced current is larger if the area of the loop is larger.
- The induced current is larger if the magnet moves perpendicular to the loop.

Magnetic Flux and Faraday's law of induction

- **Magnetic flux** through area, A:

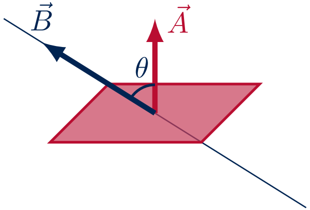
$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \int B dA \cos\theta$$

ΔV is the "induced emf" along the loop that bounds the area:

- Proportional to area of loop.
- Proportional to magnetic field.
- Proportional to rate of change of flux.
- The opposite sign if $\vec{B}$ points in the opposite direction.

Faraday's Law of induction:

$$\Delta V = -\frac{d\Phi_B}{dt}$$



Flux of magnetic field through an open surface. The potential difference in Faraday's Law is along the closed path that bounds the surface.

Magnetic Flux and Faraday's law of induction

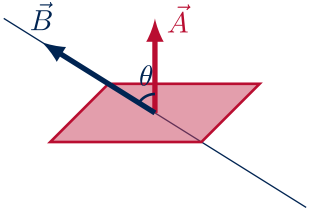
- We introduce **magnetic flux** through an area defined by a loop as:

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \int B dA \cos\theta$$

- **Faraday's Law** of induction:

$$\Delta V = -\frac{d\Phi_B}{dt}$$

ΔV is the change in potential energy per unit charge from going around the loop once.



Flux of magnetic field through a surface enclosed by a loop.

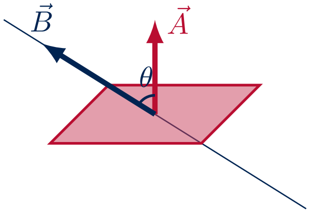
Magnetic Flux and Faraday's law of induction

- **Faraday's Law** of induction:

$$\Delta V = -\frac{d\Phi_B}{dt}$$

ΔV is the change in potential energy per unit charge from going around the loop once.

- **Lenz's Law** tells us which way the induced current goes around the loop (allowing us to ignore the minus sign):
  - If the **flux is increasing** in the loop, then the induced current creates a magnetic field that is in the **opposite direction** of the original field.
  - If the **flux is decreasing** in the loop, then the induced current creates a magnetic field that is in the **same direction** as the original field.



Flux of magnetic field through a surface enclosed by a loop.

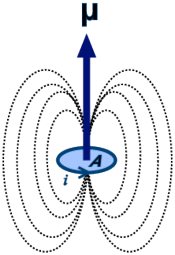
Review: Magnetic moment

- An induced current creates its own magnetic field.
- The magnetic field in the loop is aligned with the magnetic moment of the induced current in the loop:

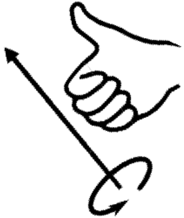
$$\vec{\mu} = I\vec{A}$$

- The loop is like a bar magnet with magnetic moment $\vec{\mu}$, thus it will experience a torque that will tend to align the magnetic moment with the magnetic field that is inducing the current:

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$



Magnetic moment for a loop of current.



Right hand rule for axial vectors.

The minus sign in Faraday's Law

- The sign of the flux depends on the choice that we make for the direction of $d\vec{A}$.

$$\Delta V = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \vec{B} \cdot d\vec{A}$$

- This will determine the sign of ΔV.
- Clearly, it's not arbitrary: If we look at current through a loop, it will flow in a given direction (from high V to low V, in the direction of $\vec{E}$).
- The minus sign is there to conserve energy!
- **It refers to the direction of the magnetic moment of the induced current relative to the (arbitrary) choice that was made for the direction of $d\vec{A}$.**

The minus sign in Faraday's Law

- The sign of the flux depends on the choice that we make for the direction of $d\vec{A}$:

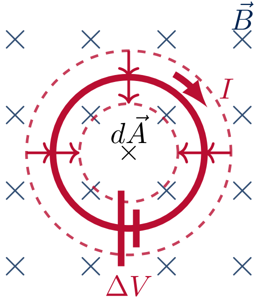
$$\Delta V = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \vec{B}\cdot d\vec{A}$$

- This will determine the sign of ΔV. **The easiest is to ignore the sign and use Lenz's Law** to figure out which way the induced current goes.

The minus sign in Faraday's Law

- However, the sign is meaningful, consider the case with the shrinking ring in a constant field:
  - If we choose $d\vec{A}$ **into the page**, then Φ_B is positive and decreasing in magnitude, so $\frac{d\Phi_B}{dt}$ **is negative**.
  - **ΔV is thus positive**.
  - With Lenz's Law, we can argue that the current is clockwise.
  - By choosing the direction of $d\vec{A}$, we define what we consider to be the positive direction of current (with our right hand). **The sign of ΔV tells us whether the magnetic moment from the induced current is in the same direction as our choice of $d\vec{A}$** .
  - The magnetic moment of the induced current is in the **same direction** as $d\vec{A}$ in this case.

$$\boxed{\Delta V = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \vec{B}\cdot d\vec{A}}$$

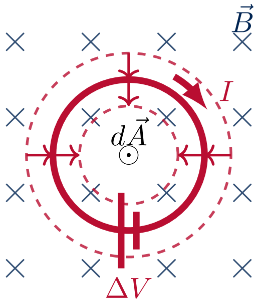


A shrinking loop in a magnetic field into the page will have clocwkise induced current.

The minus sign in Faraday's Law

- However, the sign is meaningful, consider the case with the shrinking ring in a constant field:
    - If we choose $d\vec{A}$ **out of the page**, then Φ_B is negative and increasing in value (but decreasing in magnitude), so $\frac{d\Phi_B}{dt}$ **is positive**.
    - **ΔV is thus negative**.
    - With Lenz's Law, we can argue that the current is clockwise.
    - By choosing the direction of $d\vec{A}$, we define what we consider to be the positive direction of current (with our right hand). **The sign of ΔV tells us whether the magnetic moment from the induced current is in the same direction as our choice of $d\vec{A}$.**
    - The magnetic moment of the induced current is in the **opposite** direction as $d\vec{A}$ in this case.

$$\boxed{\Delta V = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \vec{B}\cdot d\vec{A}}$$



A shrinking loop in a magnetic field into the page will have clocwkise induced current.

The minus sign in Faraday's Law

- Usually, can ignore and just use Lenz's Law (i.e. conservation of energy) to argue the direction of the induced current.
- If it's not clear, it's easiest to **pick** $d\vec{A}$ **so that** $\vec{B} \cdot d\vec{A} > 0$ (so that $d\vec{A}$ has at least a component parallel to $\vec{B}$).
- Sometimes there's no choice (e.g. a rotating coil), you need to pay attention and be clear about your choice of $d\vec{A}$.

Lenz’s Law

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Faraday’s Law and Lenz’s Law, and Real World Examples

Conservation of energy to the rescue!

- The current induced in a coil will create its own magnetic field.
  - Current in a coil creates a magnetic field, right?!
- The total magnetic field is the “original” magnetic field plus the one created by the induced current.

$\rightarrow$ If the magnetic field from the induced current changes the flux in a way to induce more current, then we are creating an infinite amount of current (and energy)!

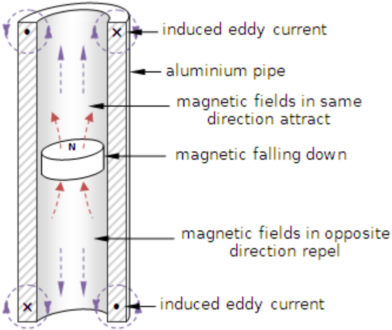
$\rightarrow$ The induced current must create a magnetic field that opposes the change in flux of the original magnetic field.

Lenz's Law

- The induced current is in such a direction as to create a magnetic field that opposes the change in flux:
    - If the **flux is increasing** in the loop, then the induced current creates a magnetic field that is in the **opposite direction** of the original field.
    - If the **flux is decreasing** in the loop, then the induced current creates a magnetic field that is in the **same direction** as the original field.

Lenz's Law

- The magnetic field flux above the magnet decreases if the magnet falls down.
  - A current is induced to create a magnetic field to restore the flux
  - That magnetic field attracts the magnet!



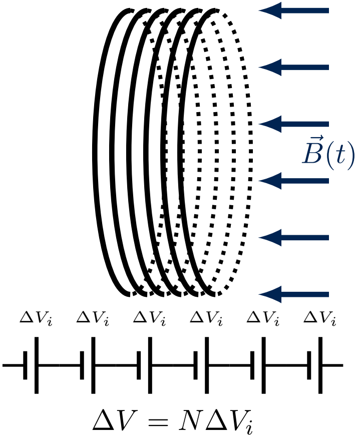
http://physicsbuzz.physicscentral.com/2014/09/build-your-own-time-warp-tube.html

Current in a coil with N loops

- Each loop can be considered separately (each has an induced emf from the flux changing through it).
- Each loop is like a little battery, and the loops are all in series.
- The (identical) emfs from each loop just add together:

$$\Delta V = -N\frac{d\Phi_B}{dt}$$

where Φ_B is the flux through one loop.



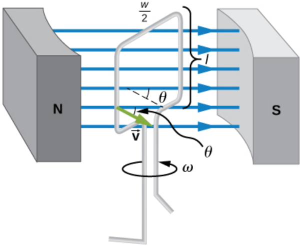
Motional emf

- Refers specifically to the case when the magnetic field is constant and the change in flux comes from motion:

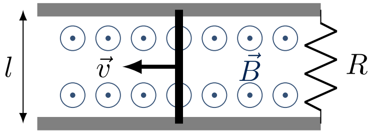
$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \int B dA \cos\theta$$

Two options:

1. The angle is changing with time (a generator).
2. The area of the loop is changing with time (the sliding bar).



Flux is changed by rotation.

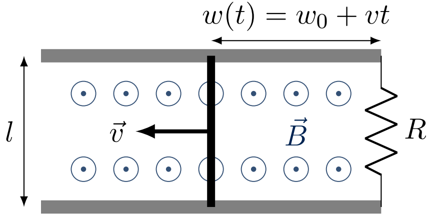


Flux is changed by change in area.

A bar moving on rails

- The area increases with time:

$$A = l(w_0 + vt)$$



A bar moving on rails.

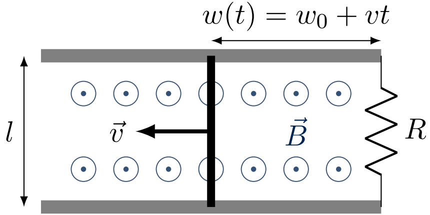
A bar moving on rails

- The area increases with time:

$$A = l(w_0 + vt)$$

- The flux is ($\vec{A}$ out of the page):

$$\Phi_B(t) = AB = Bl(w_0 + vt)$$



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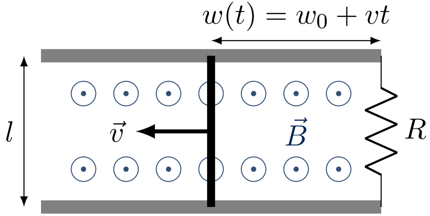
- The induced emf:

$$\Delta V = -\frac{d\Phi_B}{dt} = -Blv$$



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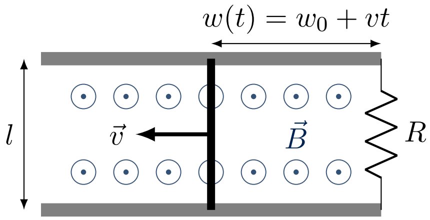
- The induced current:

$$I = \frac{\Delta V}{R} = -\frac{Blv}{R}$$

(negative, $\vec{\mu}$ into the page since $\vec{A}$ out of the page, clockwise current)



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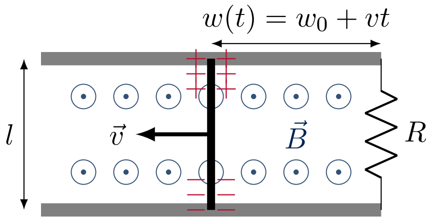


Remember the Hall Effect?

- Free charges feel a magnetic force:

$$F_B = qvB$$

- Equilibrium between the electric and magnetic forces:

$$qvB = qE = q\frac{\Delta V}{l} \quad \rightarrow \quad \Delta V = Blv$$

The emf exists event if there is no closed loop or rails!

A bar moving on rails

- The induced current:

$$I = \frac{\Delta V}{R} = -\frac{Blv}{R}$$

- The magnitude of the magnetic force on the bar is:

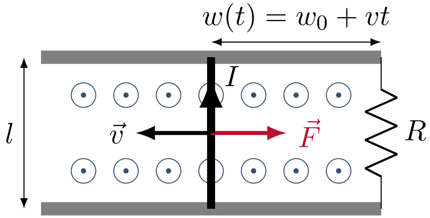
$$F = IlB = \frac{B^2 l^2 v}{R}$$

- The power to move the bar is (force exerted at constant speed):

$$P = Fv = \frac{B^2 l^2 v^2}{R}$$

- The power dissipated in the resistor is:

$$P = I^2 R = \frac{B^2 l^2 v^2}{R}$$



There is a force on the bar from the induced current

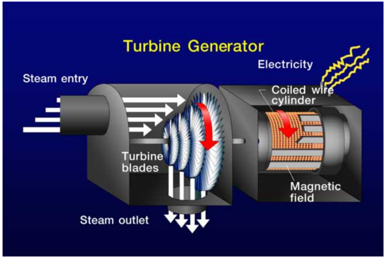
Conservation of energy

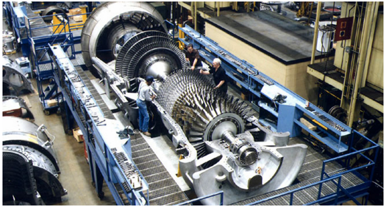
The force has to oppose the motion, or you would get free current!
The rate at which work is done to pull the bar is the same as the rate at which energy is dissipated in the resistor.

Real World Examples

The Generator

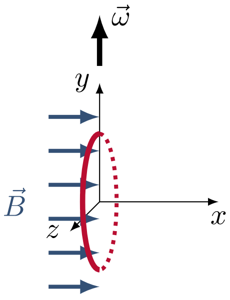
- Convert mechanical work into electrical energy.
- You have to do work (exert a torque) to keep a coil spinning in a magnetic field.
- The work is usually done by a turbine.
- The turbine is powered by:
    - High pressure water from a change in height (hydro power).
    - High pressure steam created from a heat source (nuclear fission, fossil fuel burning).
- Wind and tides can also be used to rotate a coil in a magnetic field.



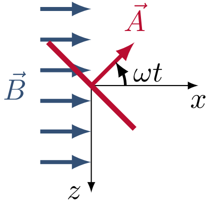


Modelling a generator

- We rotate a coil about the y axis with an angular velocity $\vec{\omega} = \omega\hat{y}$ in a constant magnetic field, $\vec{B} = B\hat{x}$.



Side view at time $t = 0$.

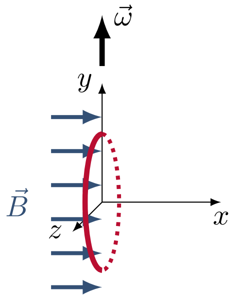


Top view at time t.

Modelling a generator

- We rotate a coil about the y axis with an angular velocity $\vec{\omega} = \omega\hat{y}$ in a constant magnetic field, $\vec{B} = B\hat{x}$.
- We choose $\vec{A}$ to be in the $\hat{x}$ direction at $t = 0$:

$$\vec{A} = A(\cos(\omega t)\hat{x} - \sin(\omega t)\hat{z})$$



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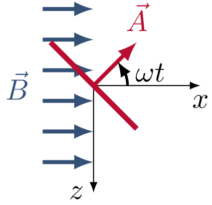

Top view at time t.

Modelling a generator

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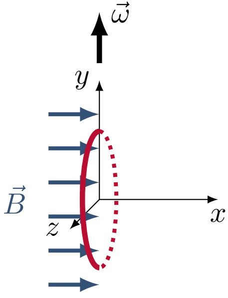
- We choose $\vec{A}$ to be in the $\hat{x}$ direction at $t = 0$:

$$\vec{A} = A(\cos(\omega t)\hat{x} - \sin(\omega t)\hat{z})$$

- The induced EMF is (for N loops in the coil):

$$\Delta V = -N\frac{d\Phi_B}{dt} = -N\frac{d}{dt}\vec{A}\cdot\vec{B} = NAB\omega\sin(\omega t)$$

This leads to sinusoidal (AC) voltage. Since ΔV changes sign as a function of time, the induced current keeps changing direction (relative direction of $\vec{A}$ and $\vec{\mu}$).



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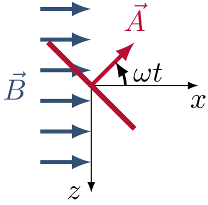

Top view at time t.

Generator: magnetic moment and torque

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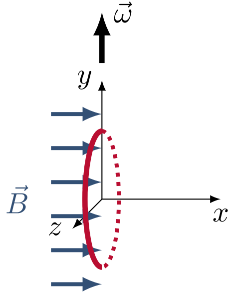
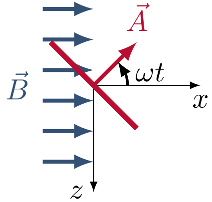


Side view at time $t = 0$.



Top view at time t.

Generator: magnetic moment and torque

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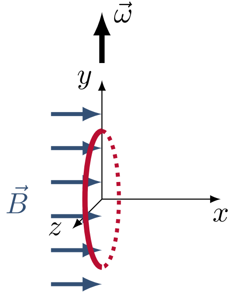
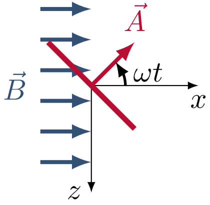
$$\Delta V = -N\frac{d\Phi_B}{dt} = -N\frac{d}{dt}\vec{A}\cdot\vec{B} = NAB\omega\sin(\omega t)$$

- If the coil has resistance R:

$$\begin{aligned}\vec{\mu} &= I\vec{A} = \frac{\Delta V}{R}\vec{A}\\ &= \frac{NAB\omega}{R}\sin(\omega t)A(\cos(\omega t)\hat{x} - \sin(\omega t)\hat{z})\end{aligned}$$



Side view at time $t = 0$.



Top view at time t.

Generator: magnetic moment and torque

- The induced EMF is (for N loops in the coil):

$$\Delta V = -N\frac{d\Phi_B}{dt} = -N\frac{d}{dt}\vec{A}\cdot\vec{B} = NAB\omega\sin(\omega t)$$

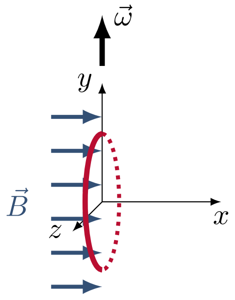
- If the coil has resistance R:

$$\begin{aligned}\vec{\mu} &= I\vec{A} = \frac{\Delta V}{R}\vec{A} \\ &= \frac{NAB\omega}{R}\sin(\omega t)A(\cos(\omega t)\hat{x} - \sin(\omega t)\hat{z})\end{aligned}$$

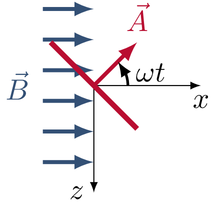
- The torque on the loop from the magnetic field is thus:

$$\vec{\tau} = \vec{\mu}\times\vec{B} = \frac{NA^2B^2\omega}{R}\sin^2(\omega t)(-\hat{y})$$

We must thus exert a torque in the positive y direction (same direction as angular velocity) opposite of the torque from the magnetic field to keep the coil rotating at constant speed.



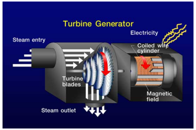
Side view at time $t = 0$.



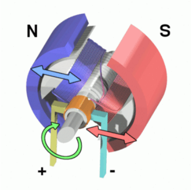
Top view at time t.

Generator or Motor?

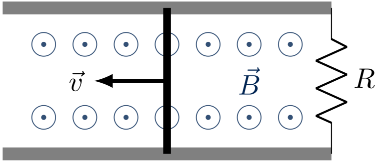
- If we do mechanical work, we can rotate a coil in a magnetic field and induce a current:
  - Mechanical energy $\rightarrow$ Electrical energy
- If we instead apply a voltage to drive a current in the coil, the coil starts to rotate (an electric motor)
  - Electrical energy $\rightarrow$ Mechanical energy
- So what about the bar on rails? If we pull on the bar, we can induce a current. What if we instead apply a voltage? What happens? What's that called?



A turbine.



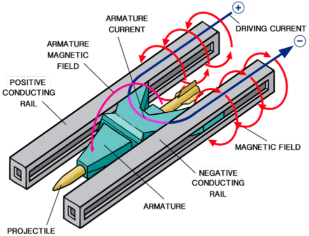
A motor.



A bar on rails.

Generator or Motor?

- If we do mechanical work, we can rotate a coil in a magnetic field and induce a current:
  - Mechanical energy $\rightarrow$ Electrical energy
- If we instead apply a voltage to drive a current in the coil, the coil starts to rotate (an electric motor)
  - Electrical energy $\rightarrow$ Mechanical energy
- So what about the bar on rails? If we pull on the bar, we can induce a current. What if we instead apply a voltage? What happens? What's that called? A railgun...



To add to the stupid things physicists have invented...

The induced electric field

- Faraday's original formulation was in terms of induced voltage:

$$\Delta V = -\frac{d\Phi_B}{dt}$$

- But we can think of this in terms of induced electric field:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

  since, after all, voltage is just (negative) work done per unit charge by an electric field.

The induced electric field

- If we want to be able to calculate an electric field from:

$$\oint \vec{E} \cdot d\vec{l}$$

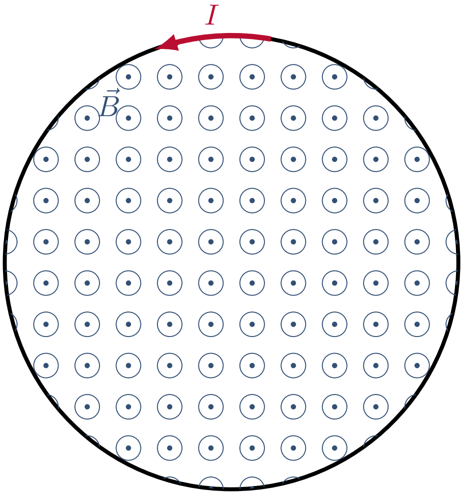
  Then, it's easier if the integral is easy...

$\rightarrow$ High degree of symmetry:
  - $\vec{E}$ constant magnitude around the loop.
  - $\vec{E}$ and $d\vec{l}$ make a constant angle between them.

- Same constraints as for Ampère's Law!

The induced electric field

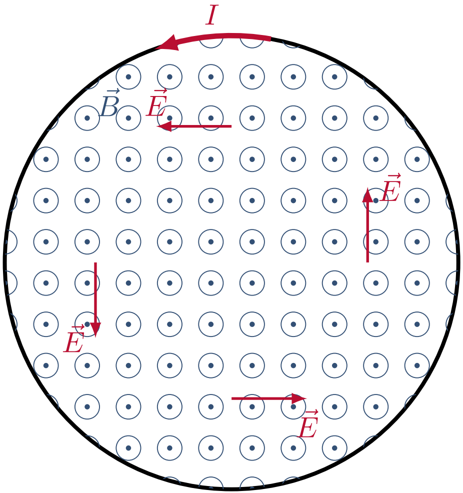
- Consider the uniform magnetic field inside of a solenoid. Assume that current is changing, so that magnetic field changes magnitude.



A time varying magnetic field induces an electric field.

The induced electric field

- Consider the uniform magnetic field inside of a solenoid. Assume that current is changing, so that magnetic field changes magnitude.
- By symmetry, we expect the electric field to make concentric circles that are coaxial with the solenoid



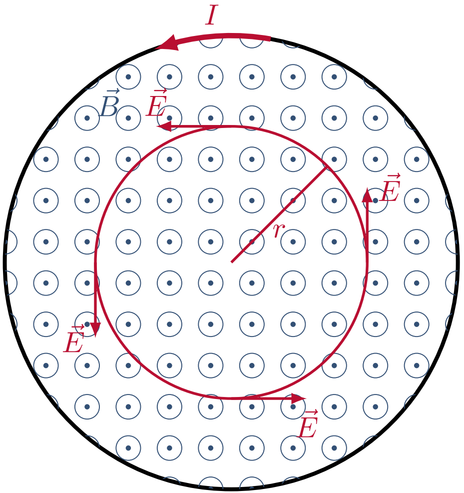
A time varying magnetic field induces an electric field.

The induced electric field

- Consider the uniform magnetic field inside of a solenoid. Assume that current is changing, so that magnetic field changes magnitude.
- By symmetry, we expect the electric field to make concentric circles that are coaxial with the solenoid
- Choose a loop over which to calculate the circulation and flux:.

$$\oint \vec{E} \cdot d\vec{l} = E(2\pi r) = -\frac{d\Phi_B}{dt}$$

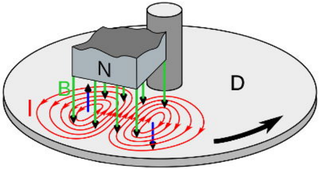
$$\therefore E = \frac{-1}{2\pi r} \frac{d\Phi_B}{dt}$$



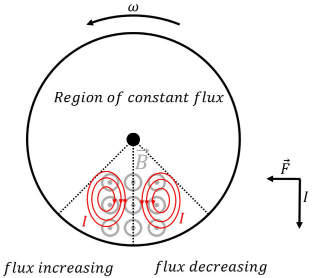
A time varying magnetic field induces an electric field.

Magnetic braking

- In the part entering the magnetic field, magnetic flux is increasing so a current is induced to create a magnetic field in the opposite direction.
- In the part exiting the magnetic field,, the magnetic flux is decreasing, so a current is induced to create a magnetic field in the same direction.
- At the middle, the currents from all loops go in the same direction, the force on those currents slow the disk down!



Magnetic brake.



Magnetic brake (note that the field is in the opposite direction compared to above!).

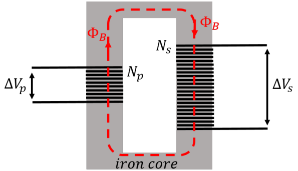
Transformers

- Used to change voltage.
- Input coil generates changing magnetic flux (AC current).
- Changing magnetic flux induces current (voltage) on output coil:

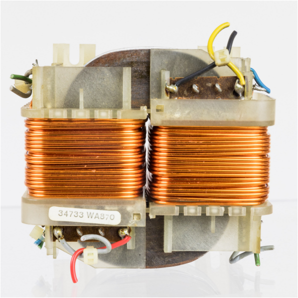
$$\Delta V_p = -N_p \frac{d\Phi_B}{dt}$$

$$\Delta V_s = -N_s \frac{d\Phi_B}{dt}$$

$$\therefore \frac{\Delta V_s}{\Delta V_p} = \frac{N_s}{N_p} \quad \rightarrow \quad \Delta V_s = \frac{N_s}{N_p} \Delta V_p$$



A transformer.



https://en.wikipedia.org/wiki/Transformer

Power lines

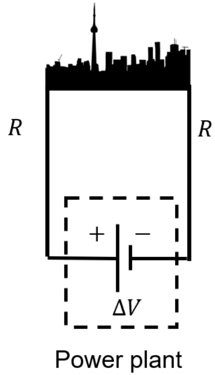
- Suppose that we wish to transmit 120kW through power lines with a resistance of $R = 0.4\,\Omega$. How much power do we loose in the lines?
- We don't know the resistance of the load (the town), but we know how much power is going through the lines (the power from the plant is used in the lines and in the town), we can thus find the current:

$$P_{plant} = I\Delta V \quad \rightarrow \quad I = \frac{P_{plant}}{\Delta V}$$

- The power dissipated in the lines is:

$$P_{line} = I^2 R_{line} = \frac{P_{plant}^2}{\Delta V^2} R_{line}$$

- If we transmit the power at $240\,$V, $P_{line} = 100\,$kW (more than 80% loss)
- If we transmit the power at $24\,000\,$V, $P_{line} = 10\,$W (much less than 1% loss)



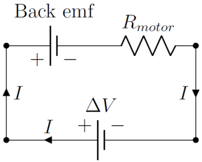
A power plant connected to a city.

Electric motor and EMF

- Recall the electric motor:
  - By running a current through a coil in a magnetic field, we could make the coil rotate
  - But now, we have a coil rotating in a magnetic field, so there must be an induced current in it, in addition to the current that we put in!

Back EMF

- In order to conserve energy, the induced current is in the opposite direction of the current driving the motor.
- The motor acts like a battery trying to oppose current coming in $\rightarrow$ “back emf”.
- When you first turn the motor on, no back emf, large current through the motor.
- As the motor spins up, the back emf grows and the current decreases.
- If there is no load on the motor, the back emf grows until it is just less than the driving emf (when it reaches the driving emf, the current is zero, the motor slows, the back emf goes down, the current goes back up $\rightarrow$ an equilibrium is reached).
- In practice there is always a load from friction.
- With no back emf, the motor would spin infinitely fast.

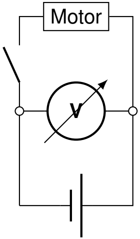


The back emf from a spinning motor acts like a battery that opposes current in a circuit. As the motor comes to speed, the back emf grows until an equlibrium is reached.

Effects of back EMF

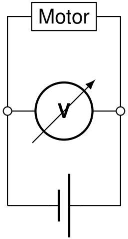
- If there is a load on the motor, it cannot spin as fast, back emf goes down, current goes up:
  - If the current is higher, the motor can do more mechanical work (I^2R).
  - If the load is too high, the current could be too high in the motor and overheat it and melt it (try blocking the air intake of your hair dryer).
- When the motor first turns on, it draws the most current:
  - When the motor in your fridge turns on, the lights in your house momentarily dim.
  - This is because the large current draw results in an appreciable voltage drop across the resistance of the electrical wires in your house.
  - Everything in the circuit gets less voltage until the motor spins up and its current goes back down.

Effects of back EMF



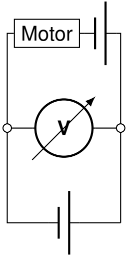
Switch open (motor off)

- No current through motor.
- Voltmeter (high resistance) measures battery voltage.
- Your lights see the same voltage as the voltmeter, they are brightest.



Switch closed (motor accelerating)

- High current through motor (low resistance).
- Voltmeter measures low voltage, since most current bypasses.
- Your lights see the same voltage as the voltmeter, they are dim.



Switch closed (motor spinning)

- Back emf from the motor reduces current in the motor.
- Voltmeter goes back up to battery voltage minus back emf voltage.
- Your lights almost as bright as before.